[Total: 20]

9 **(a)** State how a stationary wave is formed.

.....

.....

.....

..... [2]

(b) A stationary wave is set up in a stretched string. The variation of the maximum vertical displacement y of the string with distance x along the stationary wave is shown in Fig. 9.1.

P, Q and R are points on the string. The time taken by point P to travel from its maximum upward displacement to its maximum downward displacement is 9.0×10^{-3} s .

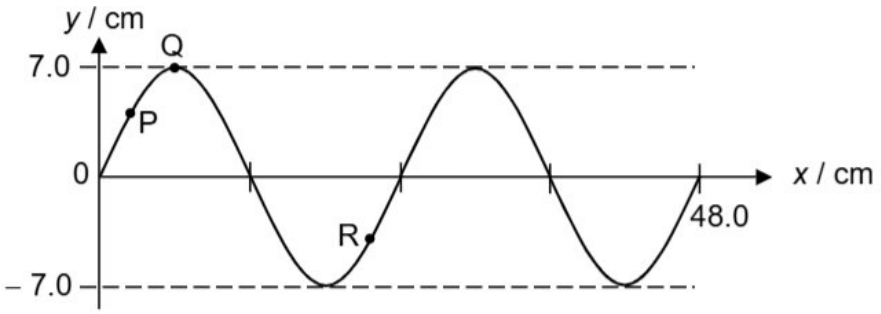


Fig. 9.1

(i) Determine, for each wave which formed the stationary wave, its

1. amplitude,

amplitude = ... cm [1]

2. frequency,

frequency = ... Hz [1]

3. speed.

speed = ... m s^{-1} [2]

(ii) State the phase difference between points

1. P and Q,

phase difference = ... rad [1]

2. Q and R.

phase difference = ... rad [1]

(c) A particle of mass *m* is trapped in a one-dimensional box of width *L* with perfectly reflecting walls. Its wavefunction is given by

$$\Psi_n(x) = \begin{cases} A\sin\left(\dfrac{2\pi x}{\lambda_n}\right) & 0 \le x \le L \\ 0 & \text{otherwise} \end{cases}$$

where λ_n is the de Broglie wavelength of the particle, *n* is the quantum number and *A* is the normalisation constant.

(i) Explain why the value of $\int_0^L |\Psi|^2 \, dx$ must be 1.

.....

.....

..... [1]

(ii) Write down an expression for λ_n in terms of L and n.

[1]

(iii) For $n = 3$, the wavefunction is $\Psi_3(x)$.

On Fig. 9.2, sketch the probability density function for the particle. Label the axes with appropriate values.

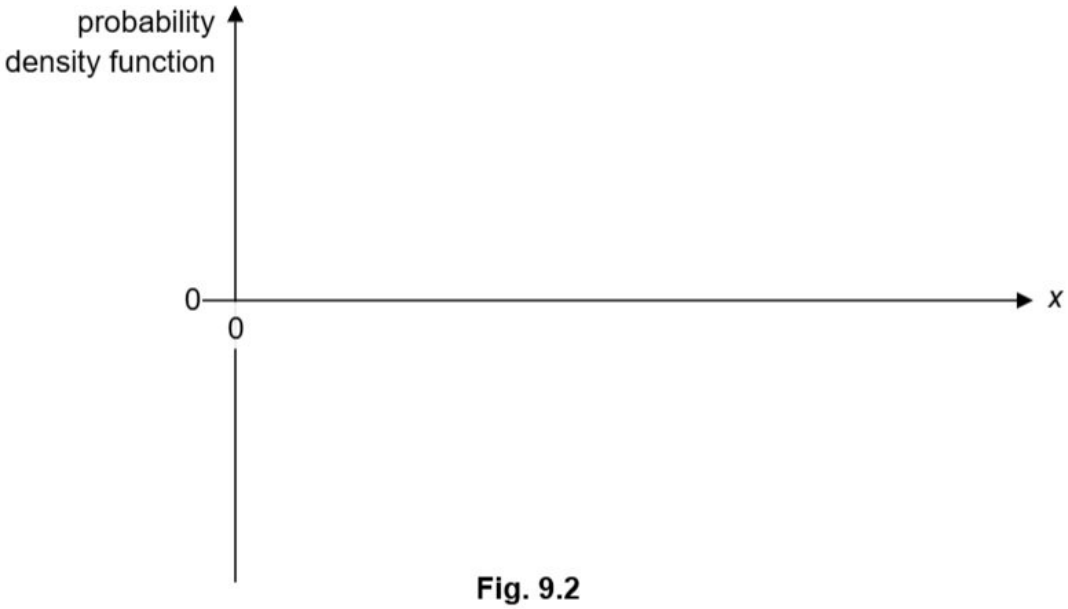


Fig. 9.2

[2]

(iv) Calculate the value of A. Show your working clearly.

[2]

(v) Suggest what might happen to the particle if the walls of the box were not perfectly reflecting.

.....

.....

..... [1]

(d) In 1980, Japanese Physicist Akira Tonomura carried out a double-slit experiment using electrons.

Fig. 9.3 shows the number of electrons that have arrived at the screen for various elapsed times *t*.

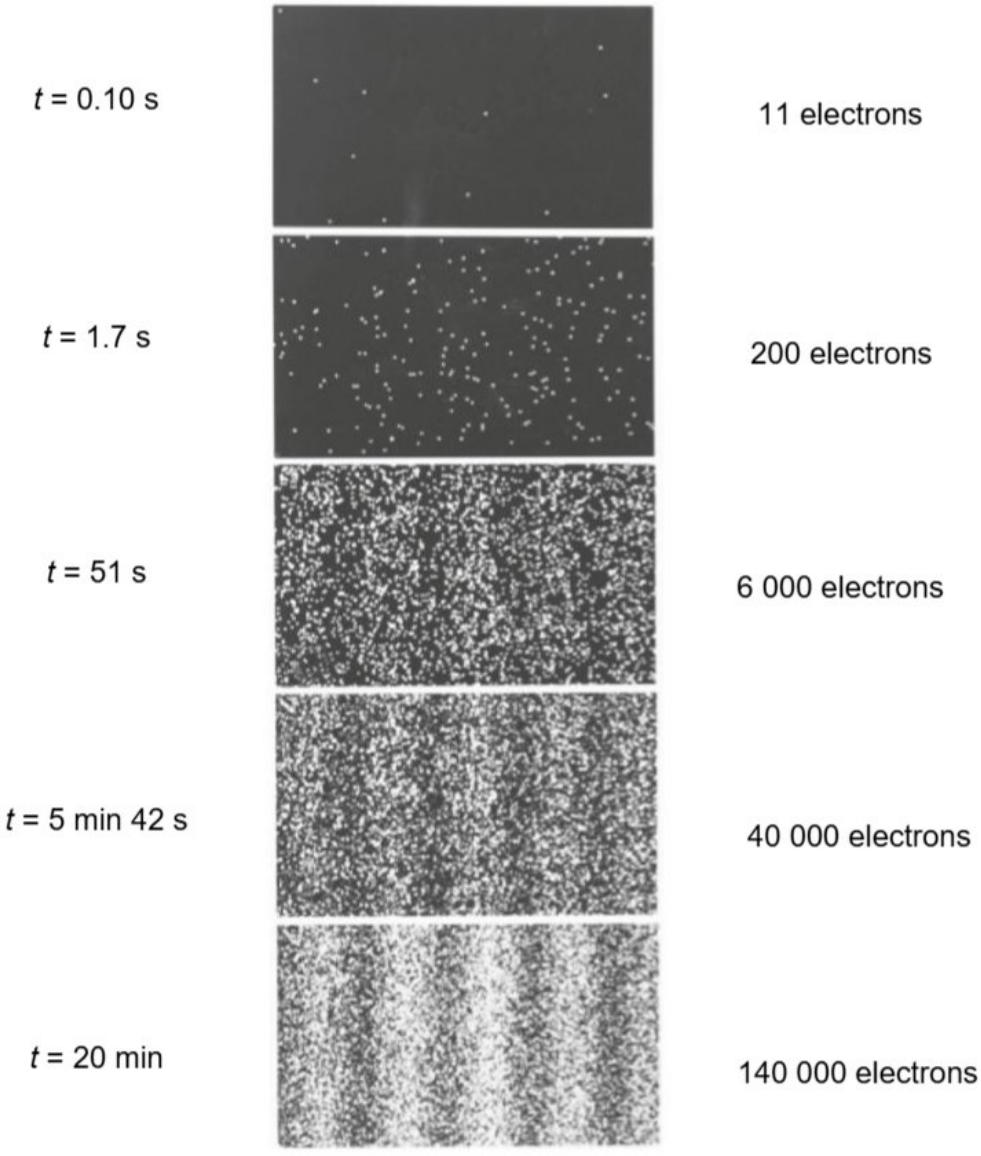


Fig. 9.3

(i) Explain how the images at various elapsed times in Fig. 9.3 show that electrons exhibit both particle-like and wave-like properties.

Particle-like: .....

.....

.....

Wave-like: .....

.....

..... [2]

(ii) The image in Fig. 9.3 shows that after an elapsed time of 20 minutes, a regular pattern has formed.

By considering the de Broglie relation, state and explain what this regularity implies about the velocities of the electrons used in the experiment.

.....

.....

.....

.....

.....

..... [3]

[Total: 20]

End of Section B